GrC-Based Image Matting for Mammogram Mass Segmentation

Project Report: MIAS and CBIS-DDSM Phases

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Base method: Hu, H., Pang, L., Shi, Z. (2016). Image matting in the perception granular deep learning. Knowledge-Based Systems, 102, 51-63.

1. PROJECT OVERVIEW

This project implements a Granular Computing (GrC)-based image matting technique, originally proposed for general foreground/background separation in natural images, and applies it to mammogram mass segmentation, first on the MIAS dataset and subsequently on CBIS-DDSM. This is a novel application: the base paper is a general-purpose matting method, not a mammography-specific technique, so there is no existing published benchmark for this exact method on either dataset. Both phases share the same five-layer architecture and five core modifications; differences between phases are driven entirely by dataset structure and annotation format, and are documented explicitly below wherever the method had to be adapted or where a genuine limitation was found.

1.1 Shared Method Architecture

LAYER	FUNCTION
Layer 1	Multi-scale CLAHE-enhanced local texture encoding (LIPW/LBPW, patch sizes 3x3 / 5x5 / 7x7)
Layer 2	Signed template matching against K-Means-learned texture templates
Layer 3	Sliding-window histogram of Layer 2 match scores
Layer 4	SVM classifier predicting mass probability per pixel
Layer 5	Matting-Laplacian alpha propagation, hybrid with intensity (gamma = 0.9 texture / 0.1 intensity)

1.2 Five Locked Modifications (both phases)

Multi-Scale CLAHE (8x8 + 16x16 tiles)

Single-Scale 5x5 baseline (ablation comparison point)

Data-driven K-Means texture templates (LBPW-derived patches, continuous signed matching)

Multi-Scale Layer 1 (3x3 + 5x5 + 7x7 concatenated)

Hybrid Alpha (gamma = 0.9 texture + 0.1 intensity, paper Eq. 33)

Implementation note (both datasets): Layer 4 uses an sklearn RBF-SVM in place of the base paper's PSVM (Probabilistic SVM as originally specified), for practical implementation reasons. This applies identically to both the MIAS and CBIS-DDSM phases.

2. MIAS PHASE: STATUS COMPLETE

2.1 Dataset

- 105 cases (CIRC 22, SPIC 19, MISC 14, NORM 50), seed = 42
- Classes ARCH / ASYM / CALC excluded (no discrete segmentable mass boundary)
- ROI extracted via radius-based crop (margin factor 0.3, capped at 150 px to prevent oversized outlier ROIs)
- Trimap generated from the same capped radius as a synthetic circle (inner = confident foreground, ring = unknown, outside = background)

- Train/test split: image-level GroupShuffleSplit over the 55 mass cases, giving 38 training and 17 test images; all 50 NORM images fall in the held-out remainder, so the saved split file lists 38 train and 67 test IDs

Known limitation, stated up front: because MIAS's mass annotations are a centroid plus radius (not a real segmentation boundary), the ground truth used for training and evaluation is a synthetic circle, not the true irregular mass shape. This structurally penalises spiculated (SPIC) masses, whose real boundary is not circular. SPIC was the weakest-performing class throughout the MIAS phase for this reason, not due to a modelling failure.

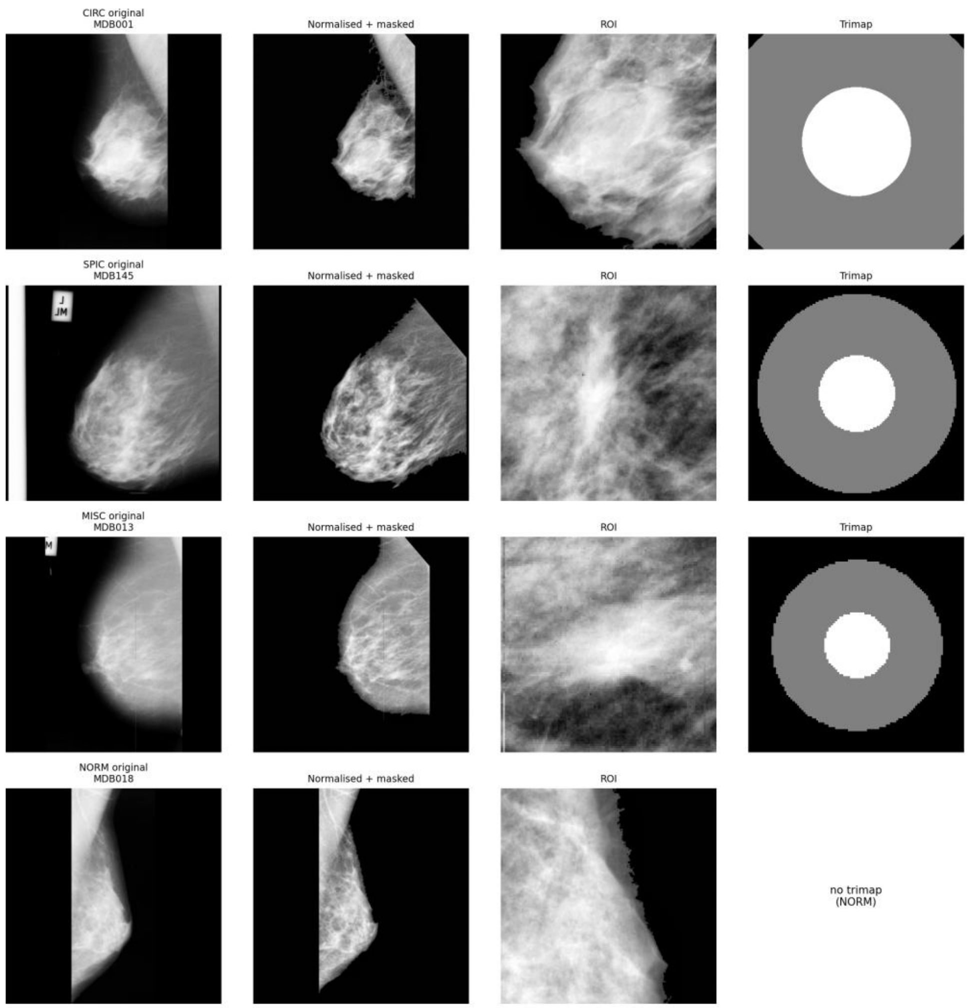

Figure 2. MIAS preprocessing verification: ROI and trimap, one sample image per class.

2.2 Key Fixes During Development

ISSUE	FIX
Filename case mismatch (mdb001.pgm vs MDB001)	.lower() normalization
Info.txt path resolution	Path built relative to parent of images directory
Bilateral duplicate cases (same image, two breast sides)	drop_duplicates(subset='image_id')
Oversized ROI (MDB001, radius 4 sd above mean)	Radius capped at 150 px, applied consistently to both ROI and trimap
Layer 1 batch too slow (60 to 90 min projected)	Vectorized via numpy.lib.stride_tricks.sliding_window_view, reduced to under 2 min
Histogram integral image off-by-one	Prepended zero row/column before cumsum

ISSUE	FIX
K-Means template diversity checks used cosine similarity	Switched to Euclidean distance
Templates collapsed to 2 or 3 dominant clusters at K = 16	Switched K-Means input from raw intensity to LBPW-derived (texture-differential) patches
K selected via silhouette score (wrong criterion, picked K = 4, worst actual discrimination)	K selection re-based on actual mass-vs-background discrimination, not clustering compactness
Layer 2 matcher used a bucketed high/medium/low threshold, discarding template identity in medium band	Rewritten as continuous signed-weight matching
SVM GridSearchCV(n_jobs=-1) hung on Colab	Diagnosed as known Colab/joblib issue; switched to n_jobs=1
Ablation baseline arm (baseline_3x3_manual) silently produced zero results	Manual hand-designed templates were hardcoded to 8, while K had been re-selected to 10 after the matcher fix, so the SVM expected 10-wide input and got 8-wide, and every row was silently skipped. Fixed by parametrizing manual template count to match the actual selected K

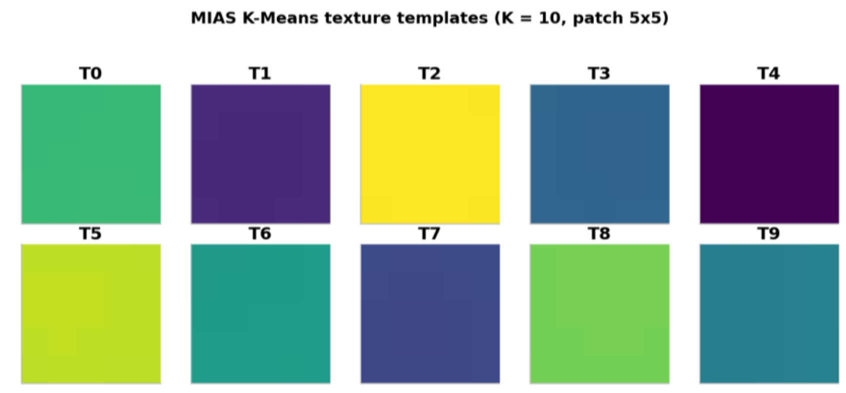

Figure 3. MIAS K-Means texture template atlas at the selected K = 10.

2.3 Locked Design Decisions

- K = 10 (final), chosen via discrimination testing, not silhouette score. The pre-fix bucketed matcher could not distinguish K = 6 / 8 / 10 at all, so the earlier K = 8 was an artifact of a broken measurement rather than a real choice; re-run under the corrected signed matcher, K = 10 was confirmed as the genuine best.
- All metrics reported per class, never as a single aggregate average.
- SPIC's negative/weak Layer 2 discrimination accepted as a genuine, structural finding. It motivates Modification 5 (hybrid alpha) as necessary, not optional.

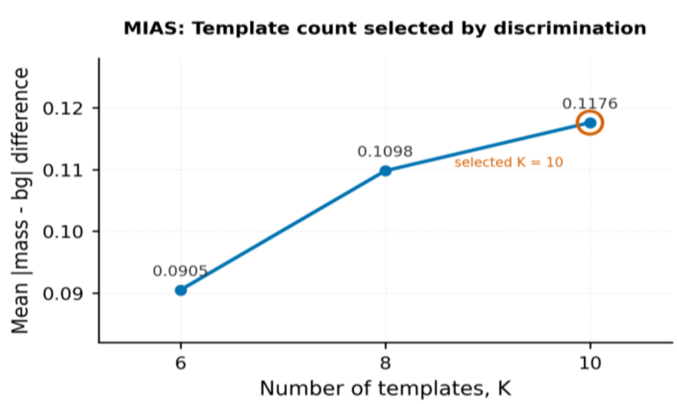

Figure 4. MIAS K-selection: mean absolute mass-versus-background score difference against the number of templates K, measured with the corrected signed matcher.

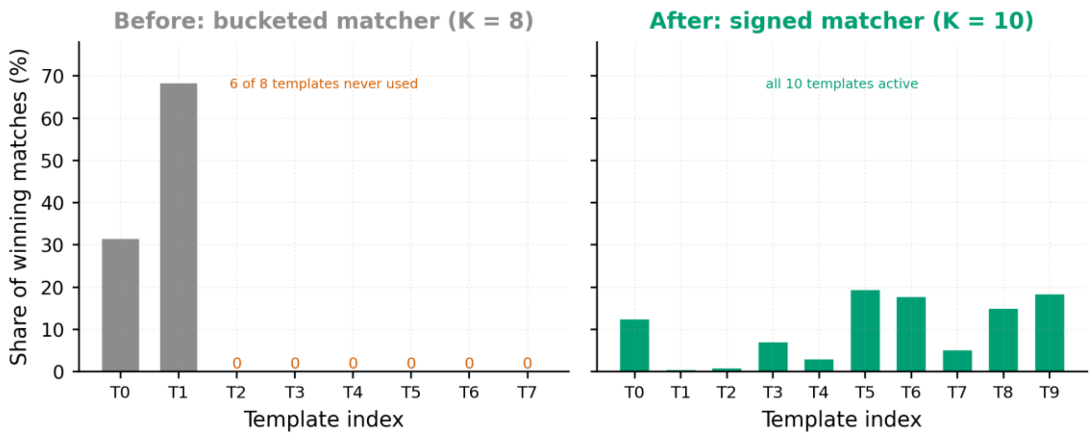

Figure 5. MIAS template utilisation before and after the Layer 2 matcher fix.

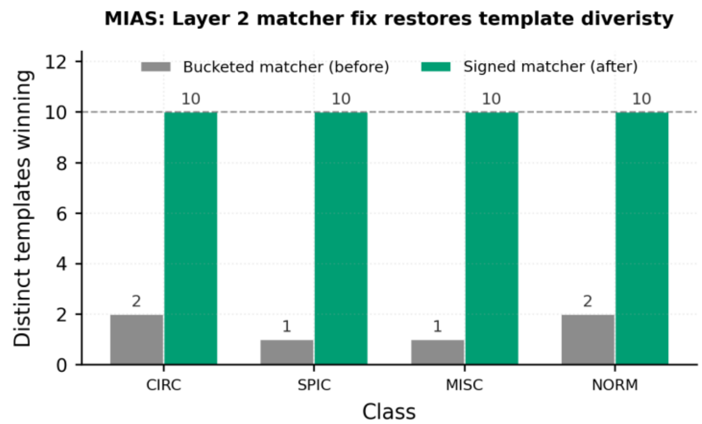

Figure 6. MIAS Layer 2 matcher fix: distinct templates winning per class, before and after the rewrite.

2.4 Final Results, Per Class (n = 105 images)

CLASS	N	DICE	SENSITIVITY	SPECIFICITY
CIRC	22	0.8416	0.7464	0.9704
MISC	14	0.8181	0.7062	0.9924
SPIC	19	0.8099	0.7019	0.9749
NORM	50	not defined	not defined	1.0000

NORM has no mass, so Dice and Sensitivity are undefined by construction, not zero.

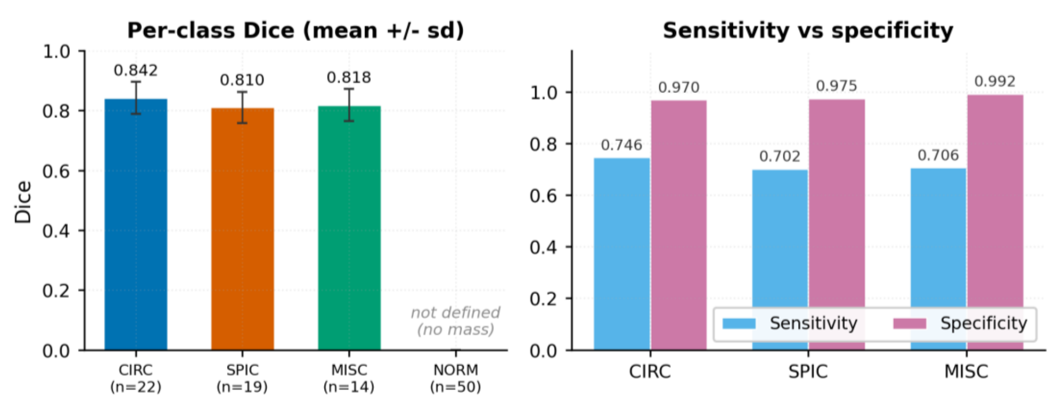

Figure 7. MIAS final per-class results: Dice with standard deviation, alongside sensitivity and specificity.

2.4b Per-Tissue-Density Breakdown

TISSUE	N	DICE	SENSITIVITY	SPECIFICITY	HD95
D (dense)	12	0.8408	0.7378	0.9896	16.57
F (fatty)	23	0.8324	0.7340	0.9796	14.31
G (fatty-glandular)	20	0.8061	0.6954	0.9681	17.64

Dense-tissue (D) cases retain the highest Dice, consistent with the well-known finding that denser tissue provides a stronger local contrast cue for the base texture features. This foreshadows the CBIS-DDSM texture/intensity distinction discussed in Section 3.5.

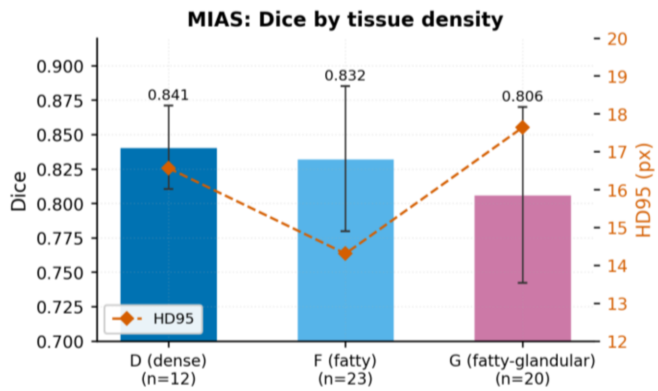

Figure 8. MIAS Dice by tissue density, with HD95 overlaid.

2.5 Ablation Study (n = 32, stratified, 8 per class)

ARM	DICE	SENSITIVITY	SPECIFICITY	HD95	ALPHA MAE
baseline_3x3_manual	0.8072	0.6941	0.9832	15.78	0.1199
mod2_5x5_single_scale	0.8114	0.7029	0.9833	16.51	0.1185
mod1234_no_hybrid	0.8110	0.7026	0.9833	16.49	0.1184
full_method	**0.8142**	**0.7078**	**0.9837**	16.31	0.1517

Dice climbs monotonically from baseline to full_method. Each modification adds incremental value, and the full hybrid method wins. full_method's higher Alpha MAE despite the best Dice is expected, it produces genuinely soft, gradient-like alpha at the mass boundary, which is penalised by MAE against a hard binary ground-truth disc that by construction cannot represent a soft boundary.

2.6 SVM (Layer 4)

- Train/test split: 38 training and 17 test images (image-level, mass cases only)
- Balanced pixel sampling: 22,800 training and 10,200 test feature rows, equal mass and background counts
- Test accuracy: 0.6279 (background recall 0.5616, mass recall 0.6943, so some bias toward predicting mass)

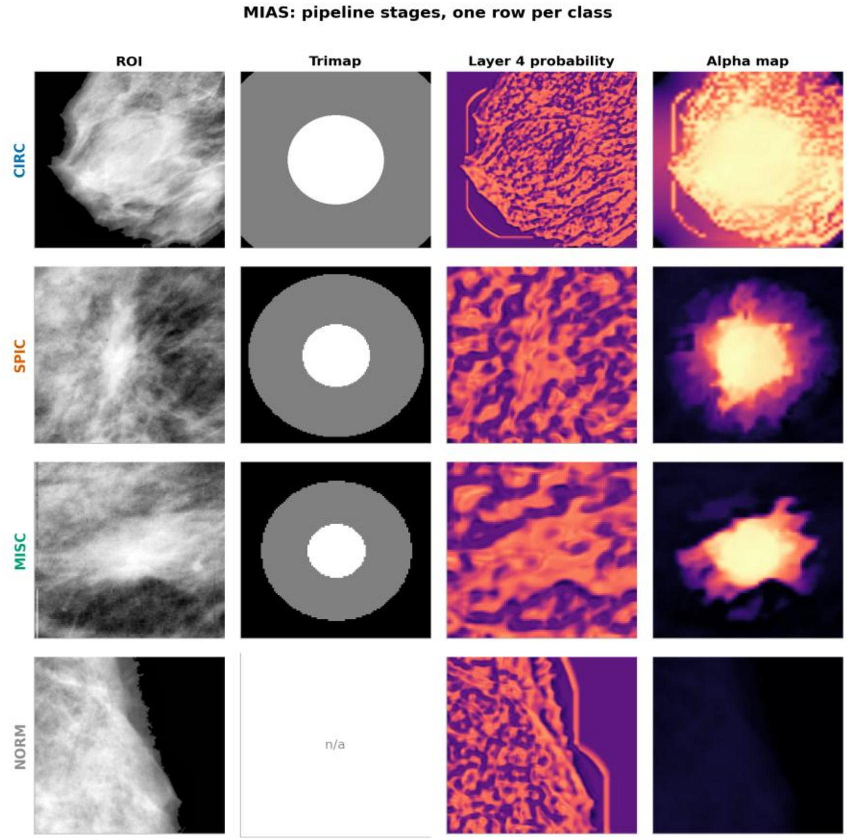

Figure 10. MIAS Section 12 sample grid: ROI, trimap, Layer 4 probability and final alpha map, one row per class (CIRC, SPIC, MISC, NORM).

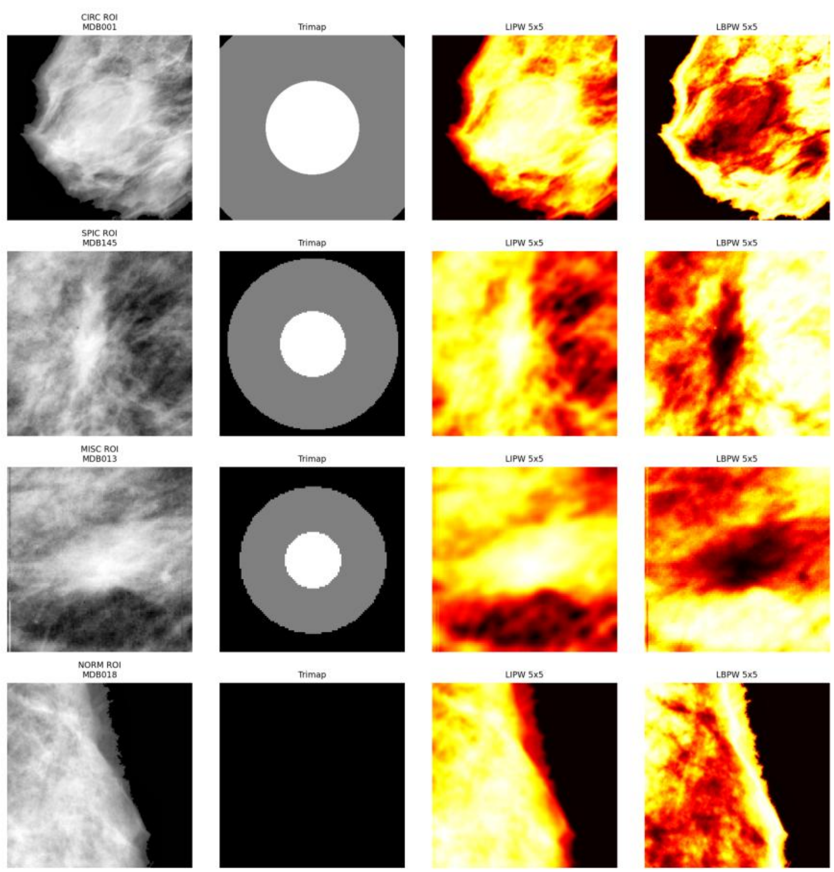

Figure 11. MIAS Layer 1 CLAHE-enhanced texture (LIPW/LBPW) sample maps.

3. CBIS-DDSM PHASE: STATUS COMPLETE

3.1 Reason for choosing CBIS-DDSM

CBIS-DDSM was chosen as a generalization test: a much larger dataset (891 mass cases against MIAS's 105) with real pixel-level segmentation masks, rather than MIAS's centroid plus radius annotation. This directly addresses MIAS's known SPIC limitation, because a real mask can represent spiculation and a circle cannot.

PROPERTY	MIAS	CBIS-DDSM
Source format	PGM (lossless)	JPEG (lossy, Kaggle mirror)
Annotation	Centroid + radius	Real pixel-level segmentation mask
Ground truth for Dice/HD95	Synthetic circle	Real radiologist mask
Classes	CIRC / SPIC / MISC / NORM	CIRC / SPIC / MISC (no NORM, see Section 3.3)
Images used	105	537
Lesions	105 (1 image per lesion)	270 (many lesions have 2 views: CC and MLO)
Train/test split unit	Image	Lesion (grouping required, see Section 3.4)

3.2 Class Mapping (locked, verified against real metadata)

Mapped from the mass margins field, pure single-label values only:

MIAS-EQUIVALENT	CBIS-DDSM VALUE	LESIONS	IMAGES USED
CIRC	CIRCUMSCRIBED	90	177
SPIC	SPICULATED	90	180
MISC	ILL_DEFINED	90	180

3.3 Scope Note: No NORM Class

CBIS-DDSM's curated release contains abnormality cases only (mass and calcification findings). The normal studies present in the original, uncurated DDSM were not carried into the curated subset. This was confirmed by direct investigation, not assumed. There is no NORM-equivalent file to source, so this phase reports 3 per-class rows, not 4. This is a scope difference from MIAS, stated explicitly here and in the results table, not a silent omission.

3.4 Data Engineering Fixes (specific to CBIS-DDSM)

ISSUE	FIX
Original CSV path resolution matched only 103 of 1696 rows (6%)	Root cause: crop and mask share the same series-UID folder in about 93% of rows, distinguishable only by content, not filename. Fixed via per-pair binary-score disambiguation (whichever of 2 candidate images is closer to binary is the mask), which resolved 1593 of 1696
Known external bug: some CBIS-DDSM mirrors have cropped-image and mask columns swapped	Explicit swap-check cell added: samples visualized, binary score computed per column, auto-corrects if needed

ISSUE	FIX
CC and MLO are two distinct images of the same lesion, not duplicates	Kept both, but train/test split grouped by lesion_key (patient, side, abnormality id), with an explicit disjoint-set leakage assertion. MIAS never needed this since it had no multi-view structure
About 32 lesions had discordant class labels between CC and MLO views	Resolved by preferring the CC view's label; discordant lesions logged for transparency
Section 11 SVM (Layer 4) initially collapsed to majority-class prediction (mass precision and recall = 0.0000)	Root cause: per-image class caps applied independently to mass and background pixels, but confident-foreground area is naturally tiny (about 0.3% of the crop), so background almost always hit the cap and mass rarely did. Fixed via per-image balanced sampling (equal counts per class) plus class_weight='balanced' and balanced_accuracy scoring as a safety net
K-selection sweep (score_k) initially reported discrimination scores far higher (0.1589) than what the trained SVM's features actually showed (about 0.001)	Root cause: score_k took the per-image maximum over K templates, a selection-biased statistic (comparable to reporting only whichever of several vague guesses happened to land per case) that inflates with K regardless of real signal. The SVM's actual training features (pooled across the dataset) showed no comparable separation

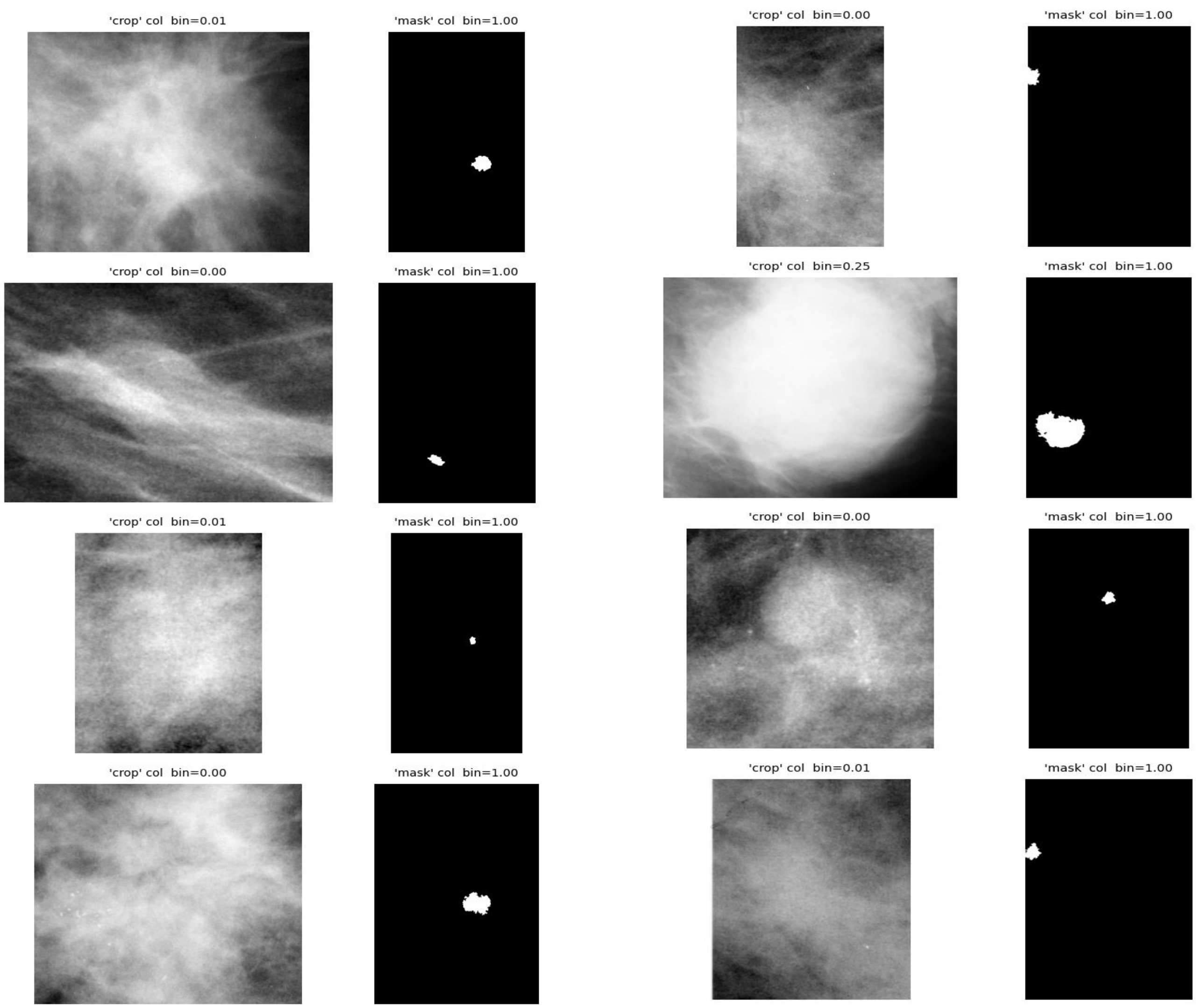

Figure 12. CBIS-DDSM swap check: candidate crop and mask side by side with their computed binary scores.

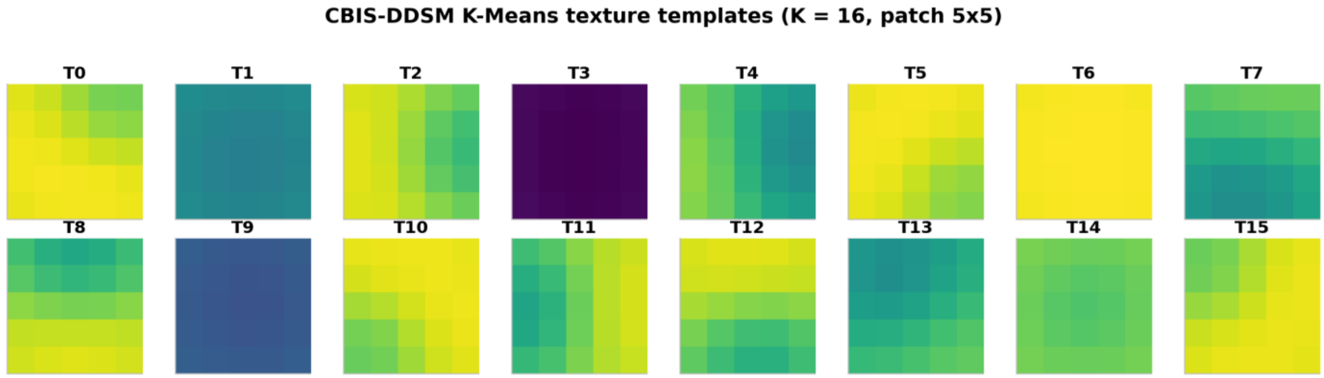

Figure 13. CBIS-DDSM K-Means texture template atlas at the selected K = 16.

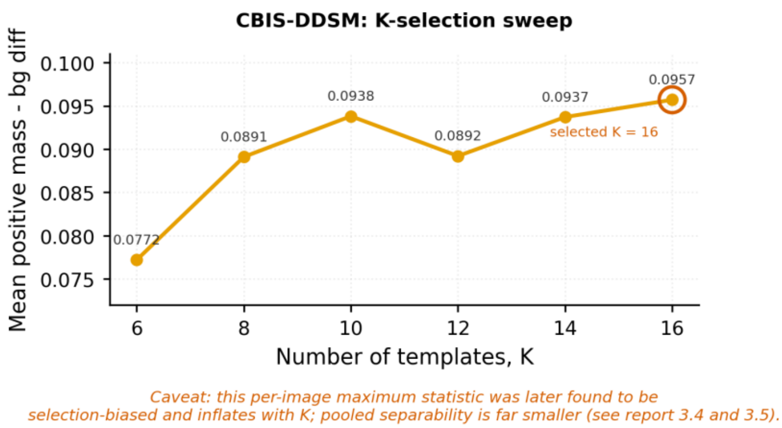

Figure 14. CBIS-DDSM K-selection sweep from K = 6 to 16. The chart also carries the caveat that this per-image maximum statistic was later found to be selection-biased.

3.5 The Core Diagnostic Finding

After ruling out the K-selection bias and a stale-cache hypothesis, direct raw-pixel-level diagnosis was performed:

FEATURE TYPE	MASS MEAN	BACKGROUND MEAN	DIFFERENCE	INTERPRETATION
Raw local texture variance (5x5)	0.022862	0.022907	-0.000045 (ratio 0.998)	No usable texture signal at all
Raw pixel intensity	0.5808 to 0.6039 (by class)	0.5509 to 0.5782 (by class)	+0.026 to +0.030, consistent across CIRC / SPIC / MISC	Real, usable signal

Conclusion: the base method's core texture features (LIPW/LBPW, K-Means templates, Layer 2 matching) are built to detect pattern and roughness, which does not differ between mass and background in this dataset at the tested patch scale. What does differ is raw brightness and density, a signal the original architecture only used as a 10% blend at the very end of Layer 5, long after Layer 4's classification was already made almost entirely from a texture channel that carries no signal.

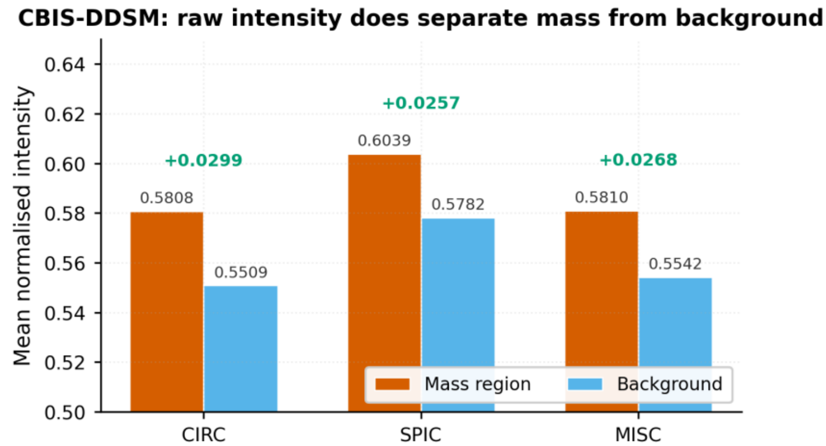

Figure 15. CBIS-DDSM per-class raw intensity diagnostic: mass against background mean intensity for CIRC, SPIC and MISC.

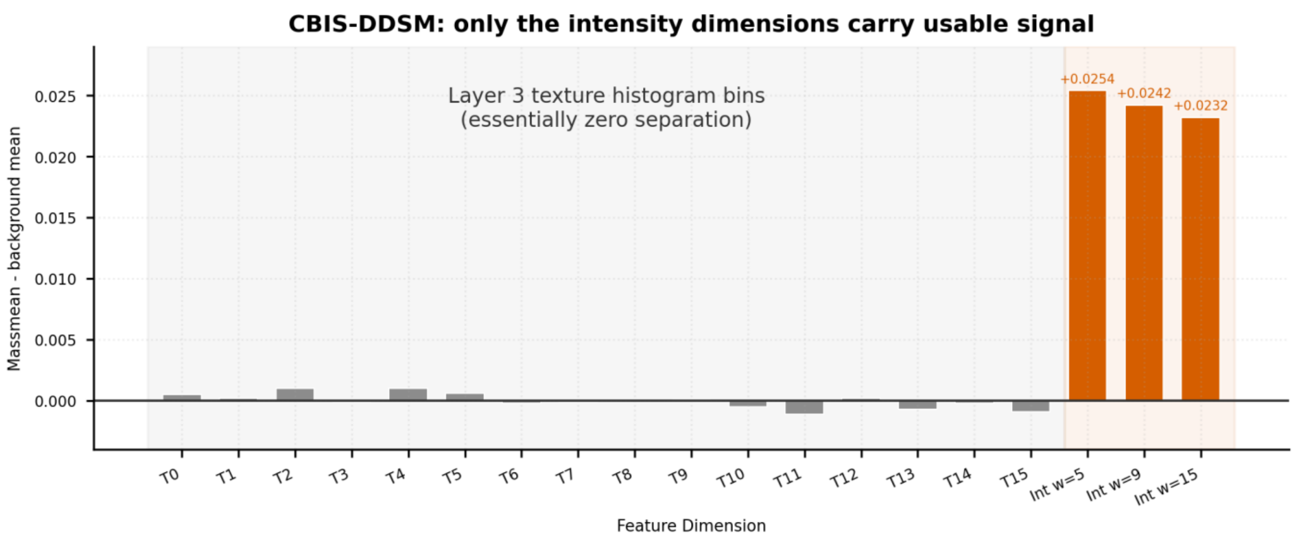

Figure 16. CBIS-DDSM per-dimension separability of the final 19-dimension feature vector. The 16 texture histogram bins separate mass from background by essentially zero, while the three intensity dimensions separate by roughly +0.023 to +0.025.

3.6 Why Increasing Dataset Size Was Not the Fix

This was tested directly, not assumed:

- CBIS-DDSM already has about 5 times more images than MIAS (537 against 105) yet performed worse pre-fix, the opposite of what a "not enough data" explanation predicts.
- A nearest-centroid baseline (the simplest possible classifier, immune to overfitting) scored below chance (0.4905) on the un-augmented feature set. More samples cannot reveal a difference between two class means that is not there; they can only estimate that non-difference more precisely.
- Three independent classifiers of very different capacity, nearest-centroid, RBF-SVM and XGBoost, all converged to the same narrow 53 to 56% band regardless of training set size or algorithm sophistication. This convergence is the signature of a feature-ceiling problem, not a data-volume or classifier-choice problem.
- A crop-tightening intervention (mirroring MIAS's tight ROI extraction, on the theory that CBIS-DDSM's loosely-padded provided crops were diluting the signal with irrelevant background) was tested directly and produced no improvement (nearest-centroid 0.4905 to 0.5110, still near chance), ruling out crop geometry as the primary cause and redirecting the investigation to the feature representation itself.

3.7 Fix Applied: Intensity-Based Features (Deviation from Locked Spec)

Given the diagnostic evidence, local mean intensity was added directly to the Layer 4 SVM's input features. This is a deliberate, documented deviation from the "5 locked modifications, unchanged from MIAS" methodology, justified entirely by the diagnostic findings above.

Model tuning and debugging progression (SVM test accuracy, Layer 4):

STAGE	TEST ACCURACY	NOTES
Original (majority-class collapse)	0.6330	Mass precision and recall = 0.0000, not a real result
Balanced training + class_weight='balanced'	0.5127	Real but weak; CV accuracy 0.5230
Nearest-centroid baseline (same features)	0.4905	Below chance, confirms no usable signal yet
After crop-tightening (0.7 margin, 40 px min, 85% area shrink)	0.5110 (nearest-centroid)	No meaningful change, crop geometry not the cause
+ single-scale intensity feature (window = 7, 17-dim)	0.5449	CV balanced accuracy 0.5454; nearest-centroid on same features 0.5340
XGBoost, same 17-dim features	0.5370	Did not beat SVM, first confirmation the bottleneck is features, not classifier
+ multi-scale intensity (windows 5/9/15, 19-dim)	**0.5593 (final)**	CV balanced accuracy 0.5516; nearest-centroid on same 19-dim features 0.5355
XGBoost, same 19-dim (multi-scale) features	0.5321	Again, did not beat SVM; the gap actually widened (2.7 points against 0.8 points at the single-scale stage), reinforcing that SVM is the better choice at both feature stages and that further feature improvements are not being left on the table by the classifier choice

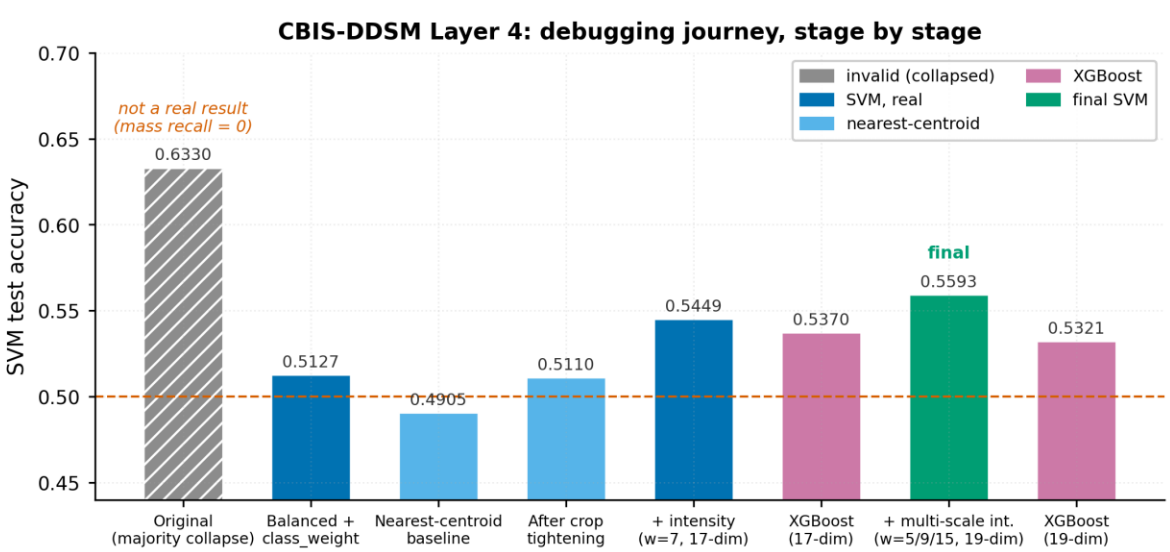

Figure 17. CBIS-DDSM Layer 4 debugging journey: SVM test accuracy at each fix stage, from the initial majority-class collapse through to the final 19-dimension result.

Refined finding on which intensity scale matters most: XGBoost per-feature importance on the final 19-dimension feature set showed the three intensity scales are not equally useful. The widest window (15 px) carried by far the most importance (0.1457), well above the medium (9 px, 0.0526) and narrow (5 px, 0.0581) scales, both of which sit in the same range as an individual texture bin (maximum single texture bin 0.0563, T11). This refines Section 3.5's conclusion: it is specifically wider-context local brightness averaging, closer to "is this region part of a generally denser structure" than "is this exact pixel bright", that carries the real signal, not intensity at any arbitrary scale.

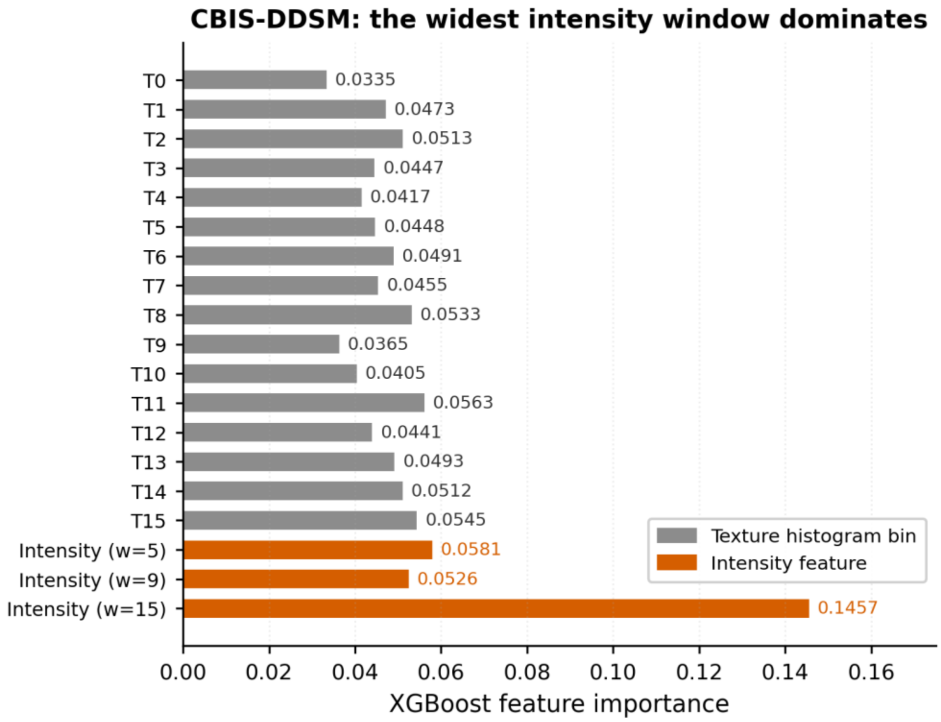

Figure 18. CBIS-DDSM XGBoost feature importance over the final 19-dimension feature set, with the three intensity windows labelled separately.

K selection (K-Means templates, Section 9):

K	DISCRIMINATION SCORE	TEMPLATES USED
6	0.0772	6 / 6
8	0.0891	8 / 8
10	0.0938	10 / 10
12	0.0892	12 / 12
14	0.0937	14 / 14
16	**0.0957**	**16 / 16**

K = 16 selected. Note, per Section 3.4, that this discrimination metric was itself later found to be inflated by a per-image selection bias. The real, pooled-level separability is much smaller, per Section 3.5.

3.8 Final Results, Per Class (n = 537 images, 270 lesions)

CLASS	N	DICE	DICE SD	SENSITIVITY	SPECIFICITY	HD95	ALPHA MAE
CIRC	177	0.6736	0.1555	0.8686	0.9853	3.2775	0.0807
SPIC	180	**0.7615**	0.1518	0.8347	0.9877	4.1928	0.0855
MISC	180	0.7146	0.1536	0.8455	0.9866	3.8574	0.0836

Key finding: SPIC is now the best-performing class, a direct reversal from MIAS (where SPIC was consistently the weakest). This is exactly the outcome predicted when moving from a synthetic circular ground truth, which cannot represent spiculation, to a real radiologist mask. It was a testable hypothesis stated at the start of this phase, and is now empirically confirmed.

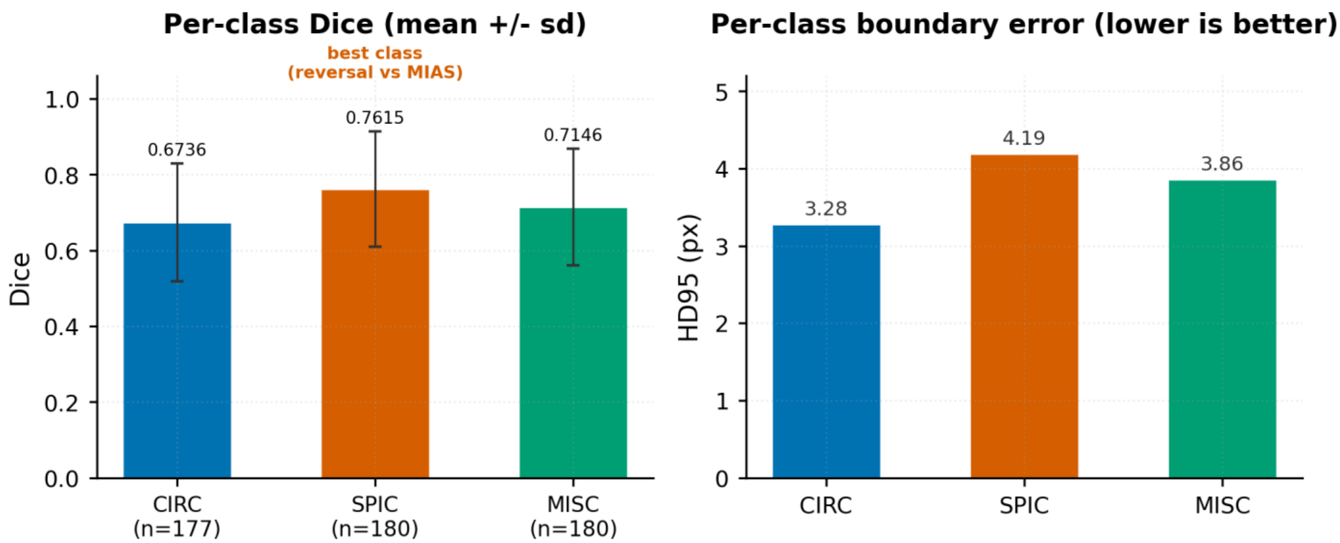

Figure 19. CBIS-DDSM final per-class results: Dice with standard deviation and per-class HD95 boundary error.

3.9 Ablation Study (n = 30, stratified, 10 per class)

ARM	DICE	SENSITIVITY	SPECIFICITY	HD95	ALPHA MAE
baseline_3x3_manual	**0.7048**	**0.8834**	0.9849	3.5095	0.0301
mod2_5x5_single_scale	0.6692	0.8401	0.9853	3.6883	0.0306
mod1234_no_hybrid	0.6681	0.8398	0.9853	3.6795	0.0307
full_method	0.6703	0.8493	0.9847	3.6537	0.0843

This reverses the MIAS finding, and it is directly explained by Section 3.5, not a contradiction of it. In MIAS, full_method (all 5 modifications) achieved the best Dice. Here, the simplest arm (baseline_3x3_manual) wins. Modifications 1 to 4 are texture-focused enhancements (multi-scale CLAHE, multi-scale Layer 1, learned K-Means templates), and Section 3.5 already established that texture carries essentially no mass-versus-background signal in this dataset. Adding more texture machinery does not help, and may add noise, when the underlying channel it operates on is not discriminative. This is a coherent, internally consistent finding across the whole diagnostic investigation, not an isolated anomaly

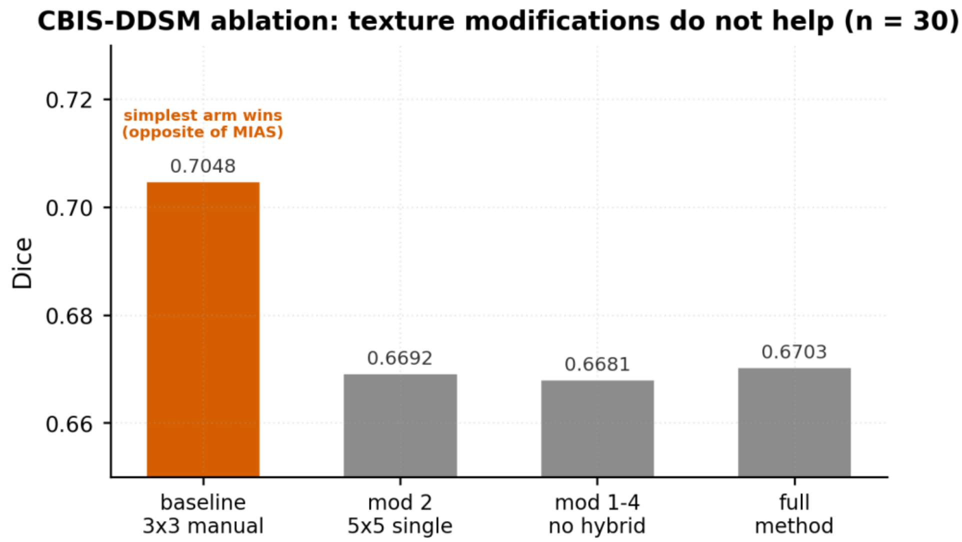

Figure 20. CBIS-DDSM ablation study: Dice by arm, with the simplest baseline arm winning, the reverse of the MIAS result.

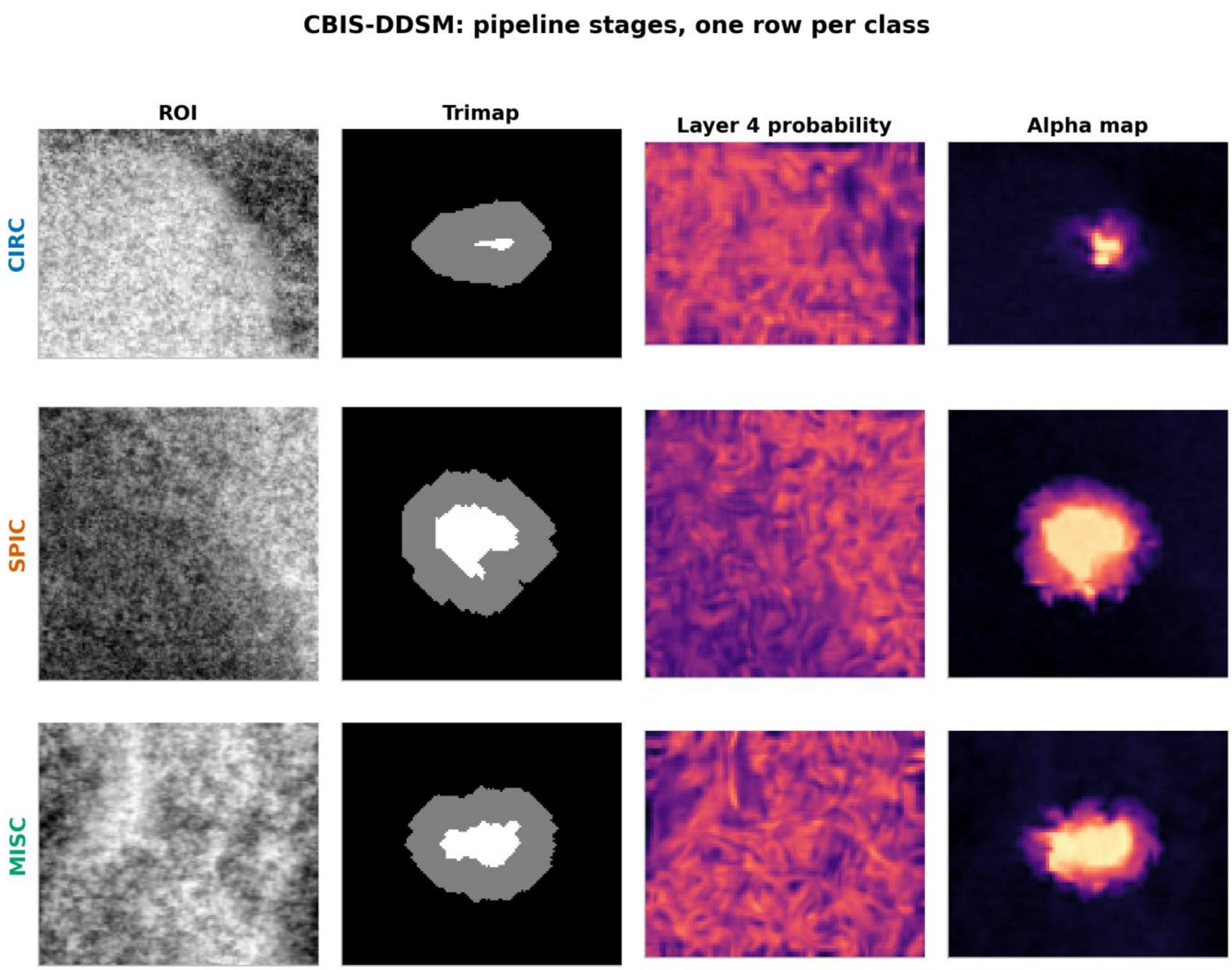

Figure 21. CBIS-DDSM Layer 5 alpha map sample grid in the final configuration, one row per class.

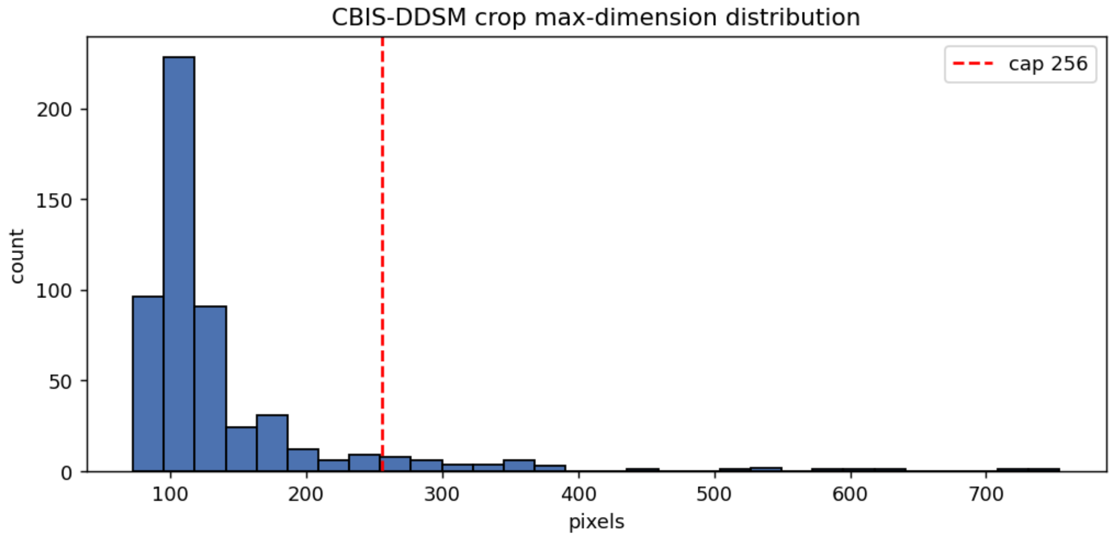

Figure 22. CBIS-DDSM crop-size distribution with the applied resolution cap, and the ROI, real mask, trimap and foreground overlay sample grid.

4. CROSS-DATASET COMPARISON

4.1 Dataset and Method Comparison

PROPERTY	MIAS	CBIS-DDSM
Images	105	537
Ground truth	Synthetic circle	Real radiologist mask
Feature basis	Texture (LBPW / K-Means) only	Texture + intensity (added)
SVM test accuracy	0.6279	0.5593
Best ablation arm	full_method (all 5 modifications)	baseline_3x3_manual (simplest)

4.2 Per-Class Dice Comparison

CLASS	MIAS DICE	CBIS-DDSM DICE
CIRC	0.8416	0.6736
SPIC	0.8099 (weakest in MIAS)	0.7615 (strongest in CBIS-DDSM)
MISC	0.8181	0.7146

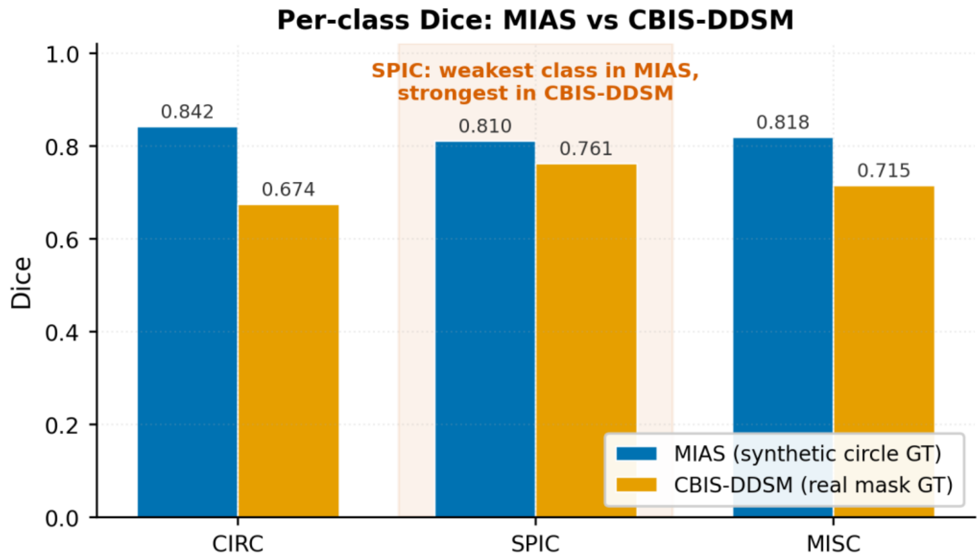

Figure 23. Per-class Dice compared across both phases, with the SPIC reversal highlighted: the weakest class under a synthetic circular ground truth becomes the strongest under a real radiologist mask.

4.3 Interpretation

MIAS's absolute Dice values are higher across the board, which is expected: MIAS's texture-based method genuinely works there because real signal exists, while CBIS-DDSM's texture channel carries almost no signal and required an architectural addition (intensity features) just to reach a modest, honestly-earned result. The one place CBIS-DDSM outperforms MIAS relatively, SPIC, is precisely where the ground-truth quality difference matters most, and is the clearest piece of evidence that the real-mask methodology is a genuine improvement for that specific class of mass shape, independent of the overall accuracy gap.

5. DEVIATIONS FROM ORIGINAL SPEC: SUMMARY FOR TRANSPARENCY

- CBIS-DDSM Layer 4 SVM input includes intensity features (single-scale, then multi-scale) in addition to the paper's texture-histogram features, justified by the diagnostic findings in Sections 3.5 and 3.6, not an arbitrary architecture change.
- CBIS-DDSM has no NORM class, a scope limitation of the curated dataset itself, not a methodological choice.
- CBIS-DDSM train/test split is lesion-grouped, not image-grouped, due to the CC/MLO multi-view structure MIAS did not have.
- JPEG against PGM was considered as a possible cause of the texture-signal absence (CBIS-DDSM's Kaggle mirror is JPEG-compressed, MIAS's source was lossless PGM). Converting the existing JPEGs to PGM would not recover already-discarded compression information, and acquiring the original uncompressed DICOM source was assessed as a large, uncertain-payoff undertaking, deferred as future work (see Section 6).